Synthesis Report

Algorithmic Analysis of Decentralized Stochastic Control with Information Delay

Comparative Evaluation: Charalambos D. Charalambous equations/model compared to other model

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Abstract

This report presents a comprehensive analysis of various algorithmic approaches for solving a decentralized stochastic control problem involving K agents, subject to an information sharing delay of $T = 1$. We benchmark an exhaustive Brute Force approach over the space of stationary strategies against an iterative Dynamic Programming (DP) algorithm (Person-by-Person) operating on a non-stationary, belief-based strategy space. The analysis empirically demonstrates the crucial importance of multi-start exploration to overcome suboptimal Nash equilibria, while highlighting the scalability limits of the purely combinatorial approach.

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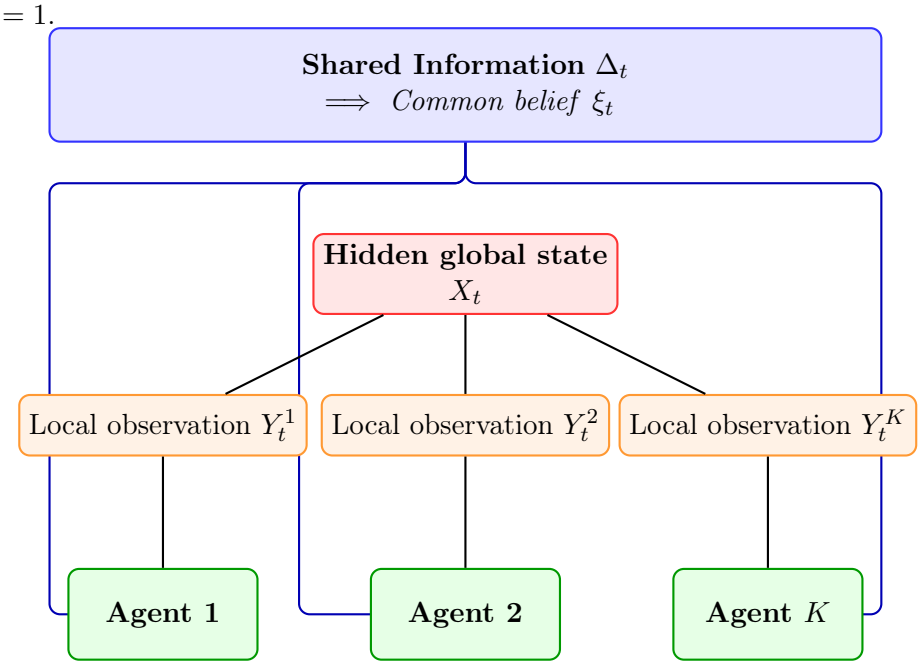
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1 Introduction and System Model

The problem studied falls within the framework of Team Theory and decentralized stochastic control. A group of K cooperative agents seeks to minimize an expected global cost over a finite horizon H.

The major difficulty lies in the **decentralized information structure**: no single agent possesses an instantaneous global view of the system state. Agents must act based on their private local observations, and they only receive the observations of other agents after a delay $T = 1$.



2 Information Structure and Theoretical Equations

2.1 Information Sets

At any given time t, the knowledge of an agent k is divided into two components:

- **Public Information (Δ_t)**: This contains the history of all observations and actions of all agents up to time $t-1$. It is common to all agents.

$$\Delta_t = \{Y_{1:t-1}^{(K)}, U_{1:t-1}^{(K)}\}$$

- **Private Information (Λ_t^k)**: This contains the local observation of agent k at the current time t.

$$\Lambda_t^k = \{Y_t^k\}$$

2.2 Common Belief and Dynamic Programming

Because the system is Markovian, the exhaustive public history Δ_t can be compressed without loss of optimality into a probability distribution over the global state, called the *belief* ξ_t:

$$\xi_t(x) = \mathbb{P}(X_t = x \mid \Delta_t)$$

The Dynamic Programming (DP) equation is then written by calculating the expectation of the instantaneous cost c_t and the future value function V_{t+1}, summing over all possible realizations of the private observations conditional on the current belief:

$$V_t(\xi_t) = \min_{\gamma_t} \sum_{y_t^{(K)}} \mathbb{P}(y_t^{(K)} \mid \xi_t) \Big[c_t(\xi_t, \gamma_t) + V_{t+1}\big(\phi(\xi_t, y_t^{(K)}, \gamma_t(y_t^{(K)}))\big) \Big]$$

where ϕ is the Bayesian belief update function. The strategy of each agent is thus a **non-stationary** function that depends on the current public belief: $\gamma_t^k(\xi_t, Y_t^k) \to U_t^k$.

3 The experience I launched

3.1 Definition of variables

$K \in \{2, 3, 4\}$ Number of agents (decision makers).
$n \in \mathbb{N}$ Time horizon; $t \in \{1, \dots, n\}$.
$T = 1$ Delay: observations/actions from step $t-1$ become *public knowledge* at step t.
$X_t \in \{0, 1\}$ Hidden state of the system at time t. Not directly observed by any agent.
$Y_t^k \in \{0, 1\}$ Noisy observation of agent k at time t. Conditionally independent given X_t.
$U_t^k \in \{0, 1\}$ Binary action taken by agent k at time t.
Δ_t Public (shared) information at time t: all past observations and actions of *all* agents, available to every agent. $\Delta_t = \{Y_{1,t-1}^k, U_{1,t-1}^k\}_{k=1}^K$.
Λ_t^k Private information of agent k at time t: only the *current* observation. $\Lambda_t^k = \{Y_t^k\}$.
I_t^k Total information of agent k at time t: $I_t^k = \Delta_t \cup \Lambda_t^k$.
$\gamma_t^k(\cdot)$ Strategy (policy) of agent k at time t: $U_t^k = \gamma_t^k(I_t^k) = \gamma_t^k(\Delta_t, Y_t^k)$.
$\xi_t^k(\cdot)$ Belief state of agent k: posterior distribution $\xi_t^k(x, \mathbf{y}^{-k}) = \mathbb{P}(X_t{=}x, Y_t^{-k}{=}\mathbf{y}^{-k} \mid I_t^k)$.
$\ell(x, \mathbf{u})$ $\mathbb{R}_{\geq 0}$ Stage cost (social): paid by all agents jointly at each step.
$V_t^k(\xi_t^k)$ $\mathbb{R}_{\geq 0}$ Value function of agent k: expected future cost from time t.

3.2 Probability Model

The hidden state is drawn at the beginning of the episode:

$$\mathbb{P}(X_1 = 0) = 0.6, \qquad \mathbb{P}(X_1 = 1) = 0.4. \tag{1}$$

3.3 Observation Model

Each agent k observes the true state X_t through a noisy binary channel. The channel is *symmetric* with agent-specific accuracy α^k:

$$\mathbb{P}(Y_t^k = y \mid X_t = x) = \begin{cases} \alpha^k & \text{if } y = x \quad \text{(correct observation)}, \\ 1 - \alpha^k & \text{if } y \neq x \quad \text{(flipped observation)}. \end{cases} \tag{2}$$

The four agents have the following sensor accuracies:

Agent k	1	2	3	4
Accuracy α^k	0.90	0.80	0.85	0.75
Error prob. $1-\alpha^k$	0.10	0.20	0.15	0.25

Example. If the true state is $X_t = 1$, Agent 1 sees $Y_t^1 = 1$ with probability 0.90, and mistakenly sees $Y_t^1 = 0$ with probability 0.10. Agent 4, the least reliable sensor, makes an error one time in four ($\alpha^4 = 0.75$).

The observations are conditionally independent given the state:

$$\mathbb{P}\big(Y_t^1{=}y^1,\ldots,Y_t^K{=}y^K \mid X_t{=}x\big) = \prod_{k=1}^{K} \mathbb{P}(Y_t^k = y^k \mid X_t = x) = \prod_{k=1}^{K} \Big[\alpha^k \cdot \mathbf{1}_{y^k=x} + (1-\alpha^k)\cdot \mathbf{1}_{y^k\neq x}\Big]. \tag{3}$$

3.4 State Transition Model

The hidden state evolves according to a controlled Markov chain. At each step, the agents produce a joint action $\mathbf{u} = (u^1, \ldots, u^K)$, and the **majority vote** of these actions determines the *target state* x^*:

$$x^* = \mathbf{1}\left[\frac{1}{K}\sum_{k=1}^{K} u^k \geq 0.5\right] \in \{0,1\}. \tag{4}$$

The transition probability then depends on whether the current state X_t already matches the target x^*:

$$\mathbb{P}(X_{t+1} = x' \mid X_t = x,\, \mathbf{U}_t = \mathbf{u}) = \begin{cases} 0.8 & \text{if } x' = x^* \text{ and } x^* = x \quad \text{(stay, voted correctly)},\\ 0.2 & \text{if } x' \neq x^* \text{ and } x^* = x \quad \text{(unintended flip)},\\ 0.7 & \text{if } x' = x^* \text{ and } x^* \neq x \quad \text{(successful push)},\\ 0.3 & \text{if } x' \neq x^* \text{ and } x^* \neq x \quad \text{(failed push)}. \end{cases} \tag{5}$$

The four transition cases are illustrated in Table 1.

Table 1: Transition probabilities for all combinations of current state X_t and majority vote target x^*.

X_t	x^*	Interpretation	$\mathbb{P}(X_{t+1}{=}0)$	$\mathbb{P}(X_{t+1}{=}1)$	Dominant outcome
0	0	Vote agrees with current state	**0.8**	0.2	Stays at 0
0	1	Vote pushes state upward	0.3	**0.7**	Moves to 1
1	0	Vote pushes state downward	**0.7**	0.3	Moves to 0
1	1	Vote agrees with current state	0.2	**0.8**	Stays at 1

Interpretation. The joint action $\mathbf{u}$ acts as a *soft push*: when all agents agree ($\mathbf{u} = (0,\ldots,0)$ or $\mathbf{u} = (1,\ldots,1)$), the system tends to move toward (or stay at) the voted state with probability 0.7–0.8, but stochastic noise always leaves a residual probability 0.2–0.3 of an unintended transition. Disagreement among agents (mixed votes) produces intermediate behavior determined by which value of u^k is in the majority.

4 Stage Cost Function

At each time step t, the team incurs a **social (shared) cost**:

$$\ell(x,\, \mathbf{u}) \;=\; \underbrace{\sum_{k=1}^{K}(x-u^k)^2}_{\text{individual tracking error}} \;+\; \underbrace{\lambda \sum_{1\le k<j\le K}(u^k-u^j)^2}_{\text{coordination penalty, } \lambda=2} \tag{6}$$

The cost has two components:

Tracking error: each agent pays $(x-u^k)^2 \in \{0,1\}$ for deviating from the true state. An agent who correctly identifies X_t and acts accordingly ($u^k = x$) incurs zero individual error.

Coordination penalty: disagreement between agents (agents choosing different actions) is penalized by $\lambda(u^k-u^j)^2$, with coupling strength $\lambda = 2$. This term is zero when all agents act identically. *Note :* The parameter $\lambda = 2$ is chosen arbitrarily

This loss function has been choosen arbitrarily, other loss function are also available The full cost table for $K = 2$ is given below:

Table 2: Stage cost $\ell(x,(u^1,u^2))$ for $K = 2$ agents, $\lambda = 2$.

x	u^1	u^2	Tracking	Coupling	$\ell(x,(u^1,u^2))$
0	0	0	$0+0=0$	$2\cdot 0=0$	**0**
0	0	1	$0+1=1$	$2\cdot 1=2$	3
0	1	0	$1+0=1$	$2\cdot 1=2$	3
0	1	1	$1+1=2$	$2\cdot 0=0$	2
1	0	0	$1+1=2$	$2\cdot 0=0$	2
1	0	1	$1+0=1$	$2\cdot 1=2$	3
1	1	0	$0+1=1$	$2\cdot 1=2$	3
1	1	1	$0+0=0$	$2\cdot 0=0$	**0**

The *total expected cost* of a strategy profile $\boldsymbol{\gamma} = (\gamma^1,\ldots,\gamma^K)$ over the horizon n is:

$$J_n(\boldsymbol{\gamma}) \;=\; \mathbb{E}^{\boldsymbol{\gamma}}\left[\sum_{t=1}^{n}\ell(X_t,\, \gamma_t^1(I_t^1),\ldots,\gamma_t^K(I_t^K))\right]. \tag{7}$$

5 Information Structure with $T = 1$ Delay Sharing

5.1 Partition into Public and Private Information

At each time t, the information available to agent k is split into two parts:

Information Structure ($T = 1$)

$$\Delta_t \;=\; \{Y^k_{1,\ldots,t-1},\, U^k_{1,\ldots,t-1}\}_{k=1}^{K} \qquad \text{(Public known to } \textit{all} \text{ agents at time } t)$$

$$\Lambda_t^k \;=\; \{Y_t^k\} \qquad \text{(Private known only to agent } k)$$

$$I_t^k \;=\; \Delta_t \,\cup\, \Lambda_t^k \qquad \text{(Total information of agent } k)$$

The key feature of $T = 1$ delay is: at every time step t, the previous step's observations and actions of *all* agents become public. As a result, agent k knows the observations of all other agents with a delay of exactly one step.

5.2 Evolution of Information Sets

Step	Δ_t (public)	I_t^k (total for agent k)
$t=1$	$\emptyset$ (no history yet)	$I_1^k = \{Y_1^k\}$ Same as no-sharing ($T{=}\infty$).
$t=2$	$\{Y_1^k, U_1^k\}_{k=1}^K$	$\{Y_1^k, Y_2^k, U_1^k\} \cup \{Y_1^j, U_1^j\}_{j\neq k}$ **Agent k knows Y_1^j for all $j \neq k$.**
$t=3$	$\Delta_2 \cup \{Y_2^k, U_2^k\}_{k=1}^K$	$I_2^k \cup \{Y_3^k, U_2^k\} \cup \{Y_2^j, U_2^j\}_{j\neq k}$
$t \geq 2$	$\left\{Y_{1:t-1}^k, U_{1:t-1}^k\right\}_{k=1}^K$	$\left\{Y_{1:t}^k,\ U_{1:t-1}^k,\ Y_{1:t-1}^j,\ U_{1:t-1}^j\right\}_{j\neq k}$

5.3 Why This Matters

Under **no sharing** ($T = \infty$), agent k only knows its own entire history $\{Y_1^k, \ldots, Y_t^k, U_1^k, \ldots, U_{t-1}^k\}$ and must *guess* the observations and intentions of the others.

Under $T = 1$ **delay sharing**, at time $t \geq 2$, agent k knows exactly what every other agent observed and did up to $t-1$. This strictly enriches the strategy space: γ_t^k can depend on Y_{t-1}^{-k} (the others' past observations), which is impossible without sharing.

6 Belief State and Bayesian Update

6.1 Definition of the Belief

Since (X_t, Y_t^{-k}) are unknown to agent k at time t, agent k maintains a **posterior belief**:

$$\xi_t^k(x, \mathbf{y}^{-k}) \;=\; \mathbb{P}\Big(X_t = x,\ Y_t^{-k} = \mathbf{y}^{-k} \mid I_t^k\Big), \qquad x \in \{0,1\}, \quad \mathbf{y}^{-k} \in \{0,1\}^{K-1}. \tag{8}$$

Note that Y_t^{-k} — the *current* observations of all other agents — remains uncertain to agent k at time t (it will only become public at $t+1$, when it enters Δ_{t+1}).

6.2 Initial Belief

At $t = 1$, $\Delta_1 = \emptyset$, so agent k only observes Y_1^k:

$$\xi_1^k(x, \mathbf{y}^{-k}) \;\propto\; \mathbb{P}(X_1 = x) \cdot \mathbb{P}(Y_1^k = y_1^k \mid X_1 = x) \cdot \prod_{j\neq k} \mathbb{P}(Y_1^j = y^j \mid X_1 = x), \tag{9}$$

normalized so that $\sum_{x,\mathbf{y}^{-k}} \xi_1^k(x, \mathbf{y}^{-k}) = 1$.

Numerical example ($K = 2$, agent $k = 1$, observation $Y_1^1 = 1$).

Using $\mathbb{P}(X_1{=}0) = 0.6$, $\alpha^1 = 0.90$, $\alpha^2 = 0.80$:

$$\xi_1^1(0,0) = \frac{0.1 \times 0.6 \times 0.2}{Z} = \frac{0.012}{Z}, \quad \xi_1^1(0,1) = \frac{0.1 \times 0.6 \times 0.8}{Z} = \frac{0.048}{Z},$$

$$\xi_1^1(1,0) = \frac{0.9 \times 0.4 \times 0.2}{Z} = \frac{0.072}{Z}, \quad \xi_1^1(1,1) = \frac{0.9 \times 0.4 \times 0.8}{Z} = \frac{0.288}{Z},$$

with $Z = 0.012 + 0.048 + 0.072 + 0.288 = 0.420$, giving:

$$\xi_1^1 \approx (0.029,\ 0.114,\ 0.171,\ 0.686).$$

Seeing $Y_1^1 = 1$ leads agent 1 to assign high probability (68.6%) to the event $(X_1 = 1, Y_1^2 = 1)$: both the state and agent 2's observation are likely to be 1.

6.3 Belief Update for $T = 1$ (Key Mechanism)

Going from t to $t+1$, two pieces of new information arrive:

Y_{t+1}^k: the **new private observation** of agent k (always private, the very latest measurement).

Y_t^{-k}: the **revealed observations** of all other agents at step t, which just became public in Δ_{t+1}. At t, these were uncertain; at $t+1$, they are *known exactly.*

This is the fundamental difference from the no-sharing case ($T = \infty$), where only item 1 arrives. The update operator $\mathcal{T}_{t+1}^k$ implements:

$$\begin{aligned}\xi_{t+1}^k(x',\, \mathbf{y}'^{-k}) \;\propto\; & \mathbb{P}(Y_{t+1}^k = y_{t+1}^k \mid X_{t+1} = x') \\ & \cdot \prod_{j \neq k} \mathbb{P}(Y_{t+1}^j = y'^j \mid X_{t+1} = x') \\ & \cdot \sum_x S_{t+1}(x' \mid x, \mathbf{u}_t) \cdot \xi_t^k(x,\, \mathbf{y}_{t,\mathrm{rev}}^{-k}),\end{aligned} \tag{10}$$

where $\mathbf{y}_{t,\mathrm{revealed}}^{-k}$ is the exact value of Y_t^{-k} now disclosed in Δ_{t+1}, and S_{t+1} is the transition kernel from equation (5).

7 Optimality Criterion and Bellman Equations

7.1 Person-by-Person (PbP) Optimality

Finding a strategy profile that *jointly* minimizes J_n is generally intractable in the decentralized setting. Instead, we seek a **Person-by-Person (PbP) optimal** profile $(\gamma^{1,\circ}, \ldots, \gamma^{K,\circ})$, which is a Nash equilibrium: no agent can reduce the total cost by unilaterally changing its own strategy. Definition PbP Optimality (Nash Equilibrium) : A strategy profile $\boldsymbol{\gamma}^\circ = (\gamma^{1,\circ}, \ldots, \gamma^{K,\circ})$ is *PbP optimal* if, for every agent k and every alternative strategy γ^k:

$$J_n(\gamma^{k,\circ},\, \boldsymbol{\gamma}^{-k,\circ}) \;\leq\; J_n(\gamma^k,\, \boldsymbol{\gamma}^{-k,\circ}), \tag{11}$$

where $\boldsymbol{\gamma}^{-k,\circ}$ denotes the strategies of all agents other than k, held fixed at their optimal values.

7.2 The New DP Equations (Bambos DP)

Given fixed strategies $\boldsymbol{\gamma}^{-k,\circ}$ of all other agents, agent k solves the following **individual Bellman equation** (this is the main theoretical contribution of the approach):

Terminal condition:

$$V_n^k(\xi_n^k) = 0. \tag{12}$$

Recursive step (for $t = n-1, \ldots, 1$):

$$\begin{aligned}V_t^k(\xi_t^k) = \inf_{u_t^k \in \{0,1\}} \Bigg[& \underbrace{\sum_{x \in \{0,1\}} \; \sum_{\mathbf{y}^{-k} \in \{0,1\}^{K-1}} \ell\Big(x,\, u_t^k,\, \boldsymbol{\gamma}_t^{-k,\circ}(\Delta_t, \mathbf{y}^{-k})\Big) \cdot \xi_t^k(x, \mathbf{y}^{-k})}_{\text{(A) immediate cost under } \xi_t^k} \\ & + \underbrace{\sum_{\substack{y_{t+1}^k \in \{0,1\} \\ \mathbf{y}_t^{-k} \in \{0,1\}^{K-1}}} \mathbb{P}\Big(Y_{t+1}^k = y_{t+1}^k,\, Y_t^{-k} = \mathbf{y}_t^{-k} \;\Big|\; \xi_t^k,\, u_t^k,\, \boldsymbol{\gamma}^{-k,\circ}\Big) \cdot V_{t+1}^k\Big(\mathcal{T}_{t+1}^k\big[y_{t+1}^k, \mathbf{y}_t^{-k}, u_t^k, \boldsymbol{\gamma}^{-k,\circ}\big](\xi_t^k)\Big)}_{\text{(B) future cost, } T=1} \Bigg]\end{aligned} \tag{13}$$

Reading the equation. Term (A) evaluates the *current-step* social cost, averaged over the uncertain state X_t and the uncertain current observations Y_t^{-k} of the other agents (both integrated against the belief ξ_t^k).

Term (B) is the key novelty of the $T = 1$ approach. Unlike the no-sharing case (where only y_{t+1}^k is summed over), here the sum runs over *two* variables simultaneously:

y_{t+1}^k: the *next* private observation of agent k (unknown at t, revealed at $t+1$);

$\mathbf{y}_t^{-k}$: the *current* observations of other agents, which are uncertain at t but **become public at** $t+1$ (entering Δ_{t+1}), and therefore appear as conditioning information in the updated belief ξ_{t+1}^k.

Each branch of the double sum leads to a *different* updated belief ξ_{t+1}^k via the operator $\mathcal{T}_{t+1}^k$ (equation (10)), and hence to a potentially different optimal future action.

8 Evaluated Resolution Methodologies

To solve this problem, we compared three decentralized paradigms against a centralized lower bound.

1. **Centralized (Omniscient)**: A single coordinator instantaneously observes $\{Y_t^1, \ldots, Y_t^K\}$ (delay $T = 0$) and computes the optimal global policy. This bound is theoretically unattainable in a decentralized setting but serves as an absolute benchmark.

2. **Decentralized Brute Force (BF)**: The algorithm exhaustively enumerates the set of **pure stationary strategies** (i.e., memoryless, such as always 0, always 1 identity or inversion). Although this space is mathematically restricted compared to belief-based strategies, the BF method guarantees the discovery of the *best* Nash equilibrium within this space.

3. **DP Identity ($T = 1$)**: An iterative algorithm implementing the equations from Section 2 (optimizing via a *Person-by-Person* approach). It explores the space of **non-stationary** strategies. Initialization is done with a single naive profile (the "Identity" strategy, where each agent plays the action corresponding to their observation).

4. **DP Multi-Start ($T = 1$)**: Identical to the DP Identity, but the optimization loop is executed multiple times, exhaustively sweeping through all 4^K pure strategies as starting points. This is designed to avoid local optima.

9 Results Analysis: The Local Minimum Phenomenon

The experimental results are summarized in Table 3.

Table 3: Cost function at balance ($H = 3, T = 1$)

Agents (K)	**Centralised**	**Brute Force**	**DP Identity**	**DP Multi-Start**	**Centralised → Multi**
$K = 2$	1.1131	1.7280	2.4600	1.7280	+55%
$K = 3$	1.5190	2.5920	5.9700	2.5920	+70.6%
$K = 4$	1.9449	3.4560	12.4566	3.4560	+82.32%

Note : Brute force evaluate stationnary strategies. The DP Multi-Start evaluate non stationnary space (belief with $T = 1$). The gain shows the local minima avoidance.

Table 4: Computation time comparaison ($H = 3, T = 1$)

Agents (K)	**Centralised**	**Brute Force**	**DP Identity**	**DP Multi-Start**
$K = 2$	10 ms	3 ms	49 ms	106 ms
$K = 3$	145 ms	41 ms	826 ms	4 392 ms
$K = 4$	2 182 ms	692 ms	11 690 ms	195 838 ms

Note : The Brute Force used in those experiences is brute force of all stationary states. Time comparaison with brute force that look for non stationnary state will be seen in part 5. Times are in milliseconds (ms). The temporal explosion pour $K = 4$ shows the complexity of calculation of decentralized beliefs.

9.1 The Apparent Failure of Identity

Initially, the DP (Identity) yielded costs vastly superior to the Brute Force (up to +260% for $K = 4$). This paradoxical behavior (a more sophisticated algorithm producing worse results) is explained by the algorithmic nature of the *Person-by-Person* search:

- The algorithm searches for a fixed point (Nash Equilibrium).
- The initialization strategy (Identity) is *already* a valid Nash equilibrium (no agent can improve the cost by unilaterally deviating), despite being a **socially disastrous** equilibrium.
- Consequently, the DP Identity converged instantly (at iteration 1) into this **local minimum**.

9.2 Validation through Multi-Start

Integrating the Multi-Start approach corrects this bias. By initializing the DP from different regions of the strategy space, the algorithm proves its mathematical capability to rigorously achieve the exact same optimal cost as the Brute Force (the BF-Multi gap is 0%). The DP thus manages to escape the gravity of local potential wells to reach the global optimum available in this space.

9.3 The choice of the parameter λ of the loss function

The results show that the BF cost is the same as the decentralized DP approach. That is because the parameter λ is too strong so the best strategy is a stationnary strategy

10 Complexity Analysis and Justification of the DP Approach

To fully appreciate the contribution of the Dynamic Programming (DP) framework developed in this work, it is fundamental to quantify the theoretical complexity of an exhaustive search (Brute Force) within the space of non-stationary strategies. This space encompasses all decision policies that depend on the public information history and private observations at each time step.

10.1 Empirical Evaluation of Computing Power

Based on benchmark measurements conducted on our test machine for a stationary system with $K = 2$ agents and a horizon $H = 3$, the exhaustive enumeration algorithm evaluated 16 strategy profiles in 3.2 ms. We can thus model the empirical processing speed of our processor, denoted as V_{PC}, as follows:

$$V_{\text{PC}} = \frac{16 \text{ profiles}}{0.0032 \text{ seconds}} = 5,000 \text{ profiles} \cdot \text{s}^{-1} \tag{14}$$

10.2 Combinatorial Explosion of the Non-Stationary Space

In a Dec-MDP model with an information delay of $T = 1$, the decision rule of an agent i at time t, denoted as γ_t^i, is a mapping from the public history Δ_t and the private observation y_t^i to the action space u_t^i, such that $\gamma_t^i : \Delta_t \times y_t^i \to u_t^i$.

For the minimal textbook case of $K = 2$ agents and an horizon of $H = 3$, let us detail the evolution of the number of admissible behavioral choices for a single agent over time:

- **At $t = 0$:** The public history Δ_0 is empty. The agent has 2 possible private observations and 2 possible actions. The number of local strategies is $2^2 = 4$.
- **At $t = 1$:** The public history Δ_1 aggregates past observations and actions, yielding $|\Delta_1| = 2^K \times 2^K = 4^K = 16$ possible trajectories. Crossed with their 2 private observations, the agent faces 32 distinct situations. The number of possible decision rules rises to 2^{32}.
- **At $t = 2$:** The public history expands further, reaching a size of $|\Delta_2| = 16^K = 256$ trajectories. Associated with the private observation, the agent must define an action for 512 different contexts. The number of possible rules explodes to 2^{512}.

The complete space of non-stationary strategies for a single agent, denoted as Γ^i, is the combinatorial product of these temporal choices:

$$|\Gamma^i| = 2^2 \times 2^{32} \times 2^{512} = 2^{546} \text{ strategies} \tag{15}$$

Since the search for a Nash equilibrium is performed over the joint profiles of all agents, the total number of combinations to evaluate for the complete system is given by:

$$|\Gamma^{\text{joint}}| = \left(2^{546}\right)^2 = 2^{1092} \approx 5.42 \times 10^{328} \text{ profiles} \tag{16}$$

10.3 Intractability of Brute Force vs. Efficiency of DP

Applying the empirical execution speed of our processor (V_{PC}), the time required (T_{BF}) to complete an exact exhaustive search over this modest horizon $H = 3$ would be:

$$T_{\text{BF}} = \frac{5.42 \times 10^{328}}{5,000} \approx 1.08 \times 10^{325} \text{ seconds} \tag{17}$$

Converting this duration into macroscopic units yields:

$$T_{\text{BF}} \approx 3.43 \times 10^{317} \text{ years} \tag{18}$$

For context, this required execution time exceeds the estimated age of the observable Universe ($\approx 1.38 \times 10^{10}$ years) by more than 300 orders of magnitude. The intractability of a naive enumeration method is therefore mathematically demonstrated.

11 Conclusion

This benchmark demonstrates that in decentralized control problems, the theoretical richness of the non-stationary strategy space (via the belief) can be nullified by local resolution methods if the initialization is suboptimal.

Brute Force remains the perfect ground-truth diagnostic tool for small systems due to its exhaustiveness. However, to break through the combinatorial wall, the Dynamic Programming algorithm, combined with strategies for avoiding local minima (such as Multi-Start or simulated annealing for large K), emerges as the architecture of choice for integrating temporal variables and policies based on shared beliefs.